Name: _____

TOPIC TEST 2

Chapter 4 Integration and areas

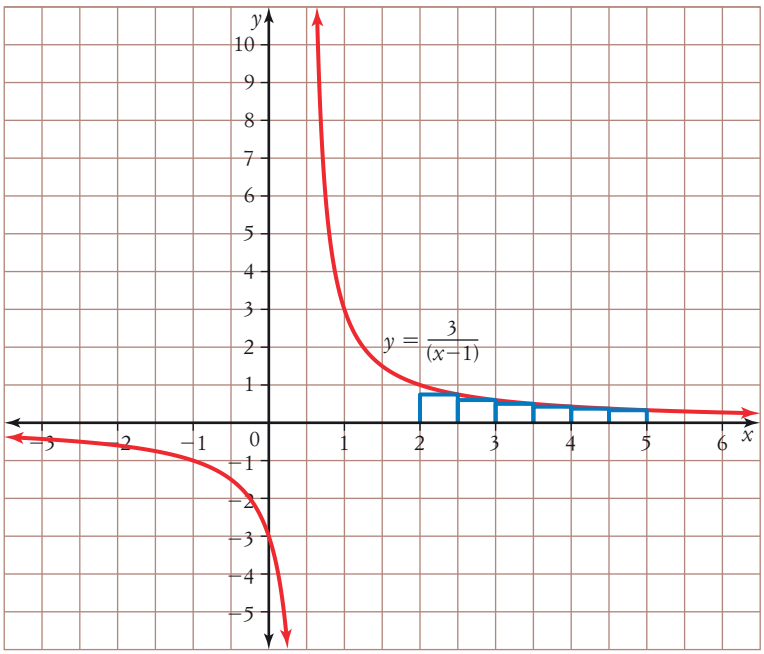
- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Five multiple-choice questions (5 marks).
- Part B: Six short-answer questions (58 marks).
- Part C: Two analysis questions (17 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Find an approximation to $\int_2^5 \frac{3}{x-1}dx$ using the rectangles below.



A $A \approx \frac{1}{2}\left(\frac{3}{2.5-1} + \frac{3}{3-1} + \frac{3}{3.5-1} + \frac{3}{4-1} + \frac{3}{4.5-1} + \frac{3}{5-1}\right)$

B $A \approx \frac{3}{2.5-1} + \frac{3}{3-1} + \frac{3}{3.5-1} + \frac{3}{4-1} + \frac{3}{4.5-1} + \frac{3}{5-1}$

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C $A \approx \frac{1}{2}\left(\frac{3}{2-1}+\frac{3}{2.5-1}+\frac{3}{3-1}+\frac{3}{3.5-1}+\frac{3}{4-1}+\frac{3}{4.5-1}\right)$

D $A \approx \frac{1}{2}\left(\frac{3}{2.25-1}+\frac{3}{3-1}+\frac{3}{2.75-1}+\frac{3}{3.25-1}+\frac{3}{3.75-1}+\frac{3}{4.25-1}\right)$

E $A \approx \frac{3}{2-1}+\frac{3}{2.5-1}+\frac{3}{3-1}+\frac{3}{3.5-1}+\frac{3}{4-1}+\frac{3}{4.5-1}$

[1 mark]

2 The area bounded by the curve $y = x^2 - 3x$ and the x-axis between $x = 1$ and $x = 3$ is given by:

A $\int_1^3 (x^2 - 3x)dx$

B $-\int_1^3 (x^2 - 3x)dx$

C $\int_1^3 (x^2 - 3x)dx - \int_0^1 (x^2 - 3x)dx$

D $\int_2^1 (x^2 - 3x)dx - \int_3^2 (x^2 - 3x)dx$

E $\int_1^2 (x^2 - 3x)\,dx - \int_2^3 (x^2 - 3x)\,dx$

[1 mark]

3 Find the definite integral $\int_0^a (2x-1)^2 dx$.

A $\frac{(2a-1)^3}{3}$

B $\frac{4a^3}{3} - \frac{4a^2}{2} + a$

C $\frac{4a^3}{3} - a$

D $(a^2 - a)^2$

E $a^2 - a$

[1 mark]

4 Find an approximation to the area under the curve $y = x^4$ between $x = 1$ and $x = 2$ by using four centred rectangles.

A $0.25 \times (1.25^4 + 1.5^4 + 1.75^4 + 2^4)$

B $0.25 \times (1^4 + 1.25^4 + 1.5^4 + 1.75^4)$

C $0.125 \times (1.125^4 + 1.375^4 + 1.625^4 + 1.875^4)$

D $0.25 \times (1.125^4 + 1.375^4 + 1.625^4 + 1.875^4)$

E $0.125 \times (1.25^4 + 1.5^4 + 1.75^4 + 2^4)$

[1 mark]

5 Simplify $\int_0^4 (2x^3 - x^2 + 5x + 4)dx - \int_0^4 (x^3 + 2x^2 - 3x - 1)dx$.

A $\int_0^4 (x^3 - 3x^2 + 8x + 5)dx$

B $\int_0^4 (x^3 - 3x^2 + 2x + 3)dx$

C $\int_0^4 (3x^3 - x^2 + 8x + 3)dx$

D $\int_0^4 (3x^3 + x^2 + 8x + 5)dx$

E $\int_0^4 (x^3 - 3x^2 + 2x + 5)dx$

[1 mark]

Part B

Short-answer questions

Six short-answer questions
58 marks
Show your working where appropriate.

1 Find the value of each definite integral:

a $\int_0^1 2x^8\,dx$

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b $\int_1^2 (6x^2 - 2x + 1)dx$

c $\int_0^3 (4x + 5)dx$

d $\int_3^5 7e^x \, dx$

e $\int_0^{\frac{\pi}{3}} \sin(x)dx$

[15 marks]

2 Find an approximation to $\int_4^6 x^2 \, dx$ by using:

a left rectangles

b right rectangles

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c centred rectangles

d a trapezium with vertical sides $f(4)$ and $f(6)$.

[15 marks]

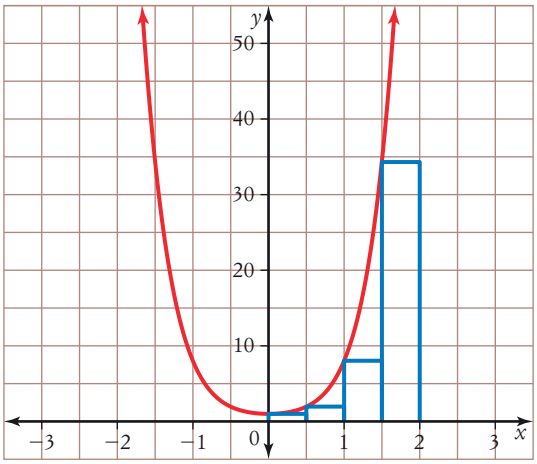
3 **a** Find the area, A, under the graph $y = 8x + 1$ between $x = 0$ and 1, 1 and 2, 2 and 3 and 3 and 4.

b Draw the graph of the area function $y = A(x)$ where $A(x)$ is the area under the curve from $x = 0$ to x using a CAS calculator and find its equation.

[15 marks]

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4 Find an approximation to $\int_0^2 (x^2 + 1)^3 \, dx$ by using the rectangles shown below:



[4 marks]

5 Without evaluating, simplify each integral.

a $-\int_3^1 e^x \, dx$

b $\int_1^2 (2x^4 - 3x^3 + x^2 - 3)dx + \int_1^2 (x^4 - x^3 + 5x^2 + 4x + 2)dx$

c $\int_{-3}^0 (2x^2 + 3)dx + \int_0^3 (2x^2 + 3)dx$

[3 marks]

6 a Evaluate $\int_0^2 (x^2 + 2x - 8)dx$.

b Find the area enclosed between the curve $y = x^2 + 2x - 8$ and the x-axis between $x = 0$ and $x = 2$.

[6 marks]

Part C

Analysis questions

Two analysis questions
17 marks
Show your working where appropriate.

1 a Evaluate $\int_1^2 (y^2 - 1)dy$.

b Evaluate $\int_0^1 (y^2 - 1)dy$.

c Sketch the graph of $x = y^2 - 1$.

d Find the area bounded by the graph $x = y^2 - 1$ and the y-axis between $y = 0$ and $y = 2$.

[11 marks]

2 The speed of a particle moving in a straight line is given by $v = 50e^t$ m/s.

a Find the exact distance the particle moves in the first 3 seconds.

b Find the distance the particle moves between the 4th and 9th seconds.

[6 marks]

[Total marks: 80]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 Each rectangle has width $\frac{1}{2}$.

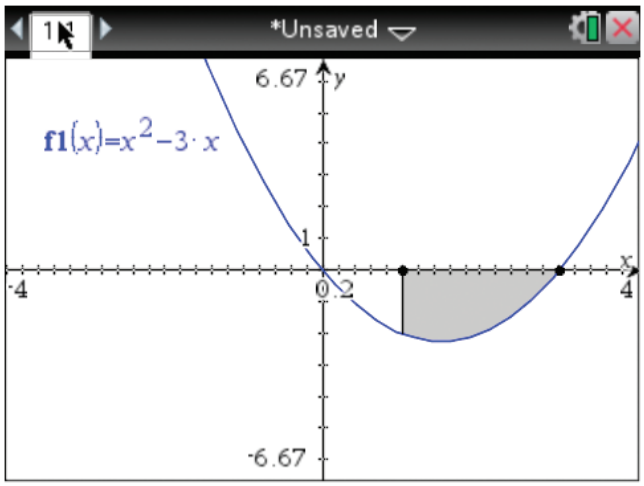
Heights are $f(2.5), f(3), f(3.5), f(4), f(4.5)$ and $f(5)$

$$A = \frac{1}{2} \times \frac{3}{2.5-1} + \frac{1}{2} \times \frac{3}{3-1} + \frac{1}{2} \times \frac{3}{3.5-1} + \frac{1}{2} \times \frac{3}{4-1} + \frac{1}{2} \times \frac{3}{4.5-1} + \frac{1}{2} \times \frac{3}{5-1}$$

$$= \frac{1}{2}\left(\frac{3}{2.5-1} + \frac{3}{3-1} + \frac{3}{3.5-1} + \frac{3}{4-1} + \frac{3}{4.5-1} + \frac{3}{5-1}\right)$$

$\therefore$ **A** **[1 mark]**

2



The algebraic area between $x = 1$ and $x = 3$ is negative, so $-\int_1^3 (x^2 - 3x)dx$ will give the physical area.

$\therefore$ **B** **[1 mark]**

3 $\int_0^a (2x-1)^2 dx = \int_0^a (4x^2 - 4x + 1)dx$

$$= \left[\frac{4x^3}{3} - \frac{4x^2}{2} + x\right]_0^a$$

$$= \frac{4a^3}{3} - \frac{4a^2}{2} + a$$

$\therefore$ **B** **[1 mark]**

4 The width of each rectangle is 0.25 units and the centres are at $x = 0.125$, 1.375, 1.625 and 1.875

Heights are $f(1.125) = 1.125^4$, $f(1.375) = 1.375^4$, $f(1.625) = 1.625^4$ and $f(1.875) = 1.875^4$

$A = 0.25 \times 1.125^4 + 0.25 \times 1.375^4 + 0.25 \times 1.625^4 + 0.25 \times 1.875^4$

$= 0.25 \times (1.125^4 + 1.375^4 + 1.625^4 + 1.875^4)$

$\therefore$ **D** **[1 mark]**

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5 $\int_0^4 (2x^3 - x^2 + 5x + 4)dx - \int_0^4 (x^3 + 2x^2 - 3x - 1)dx$

$= \int_0^4 (2x^3 - x^2 + 5x + 4 - x^3 - 2x^2 + 3x + 1)dx$

$= \int_0^4 (x^3 - 3x^2 + 8x + 5)dx$

$\therefore$ **A** **[1 mark]**

Part B – Short-answer questions

1 a $\int_0^1 2x^8\,dx = \left[\frac{2x^9}{9}\right]_0^1$ **[1 mark]**

$= \frac{2(1)^9}{9} - \frac{2(0)^9}{9}$ **[1 mark]**

$= \frac{2}{9}$ **[1 mark]**

b $\int_1^2 (6x^2 - 2x + 1)dx = \left[\frac{6x^3}{3} - \frac{2x^2}{2} + x\right]_1^2$

$= \left[2x^3 - x^2 + x\right]_1^2$ **[1 mark]**

$= [2(2)^3 - 2^2 + 2] - [2(1)^3 - 1^2 + 1]$ **[1 mark]**

$= 12$ **[1 mark]**

c $\int_0^3 (4x + 5)dx = \left[\frac{4x^2}{2} + 5x\right]_0^3$

$= \left[2x^2 + 5x\right]_0^3$ **[1 mark]**

$= [2(3)^2 + 5(3)] - [2(0)^2 + 5(0)]$ **[1 mark]**

$= 33$ **[1 mark]**

d $\int_3^5 7e^x\,dx = \left[7e^x\right]_3^5$ **[1 mark]**

$= 7e^5 - 7e^3$ **[1 mark]**

$= 7e^3(e^2 - 1)$ **[1 mark]**

e $\int_0^{\frac{\pi}{3}} \sin(x)dx = \left[-\cos(x)\right]_0^{\frac{\pi}{3}}$ **[1 mark]**

$= -\cos\left(\frac{\pi}{3}\right) - [-\cos(0)]$ **[1 mark]**

$= -\frac{1}{2} + 1$

$= \frac{1}{2}$ **[1 mark]**

2 a Width 0.5 units, heights $f(4) = 4^2, f(4.5) = 4.5^2, f(5) = 5^2, f(5.5) = 5.5^2$ **[1 mark]**

$\int_4^6 x^2\,dx \quad 0.5 \times 4^2 + 0.5 \times 4.5^2 + 0.5 \times 5^2 + 0.5 \times 5.5^2$

$= 0.5 \times (4^2 + 4.5^2 + 5^2 + 5.5^2)$ **[1 mark]**

$= 45.75$ **[1 mark]**

b Width 0.5 units, heights $f(4.5) = 4.5^2, f(5) = 5^2, f(5.5) = 5.5^2, f(6) = 6^2$ **[1 mark]**

$\int_4^6 x^2\,dx \quad 0.5 \times 4.5^2 + 0.5 \times 5^2 + 0.5 \times 5.5^2 + 0.5 \times 6^2$

$= 0.5 \times (4.5^2 + 5^2 + 5.5^2 + 6^2)$ **[1 mark]**

$= 55.75$ **[1 mark]**

c Width 0.5 units, heights $f(4.25) = 4.25^2, f(4.75) = 4.75^2, f(5.25) = 5.25^2, f(5.75) = 5.75^2$ **[1 mark]**

$\int_4^6 x^2\,dx \quad 0.5 \times 4.25^2 + 0.5 \times 4.75^2 + 0.5 \times 5.25^2 + 0.5 \times 5.75^2$

$= 0.5 \times (4.25^2 + 4.75^2 + 5.25^2 + 5.75^2)$ **[1 mark]**

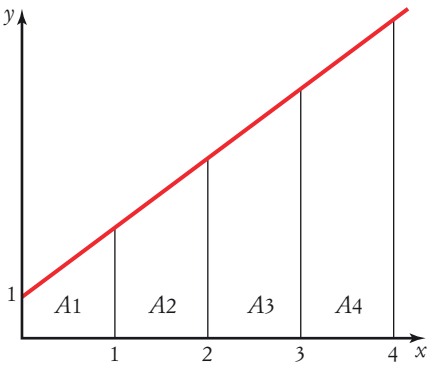
$= 50.625$ **[1 mark]**

d Area $= \frac{1}{2}h(a + b)$ where $h = 6 - 4 = 2$, $a = f(4) = 4^2 = 16$ and $b = f(6) = 6^2 = 36$ **[1 mark]**

$= \frac{1}{2} \times 2 \times (16 + 36)$ **[1 mark]**

$= 52$ **[1 mark]**

3 a



$A = \frac{1}{2}h(a + b)$

$A_1 = \frac{1}{2} \times 1 \times [(8 \times 0 + 1) + (8 \times 1 + 1)]$ **[1 mark]**

$= 5$ **[1 mark]**

$A_2 = \frac{1}{2} \times 1 \times [(8 \times 1 + 1) + (8 \times 2 + 1)]$ **[1 mark]**

$= 13$ **[1 mark]**

$A_3 = \frac{1}{2} \times 1 \times [(8 \times 2 + 1) + (8 \times 3 + 1)]$ **[1 mark]**

$= 21$ **[1 mark]**

$A_4 = \frac{1}{2} \times 1 \times [(8 \times 3 + 1) + (8 \times 4 + 1)]$ **[1 mark]**

$= 29$ **[1 mark]**

b Area at $x = 0$ is 0

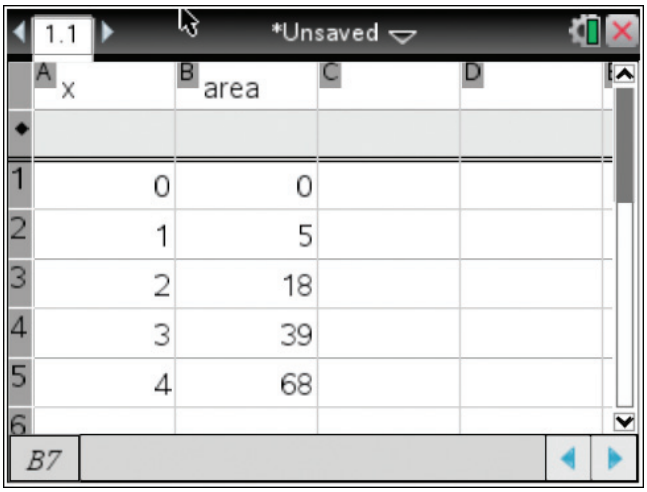
Area between $x = 0$ and $x = 1$ is $A_1 = 5$

Area between $x = 0$ and $x = 2$ is $A_1 + A_2 = 5 + 13 = 18$

Area between $x = 0$ and $x = 3$ is $A_1 + A_2 + A_3 = 5 + 13 + 21 = 39$

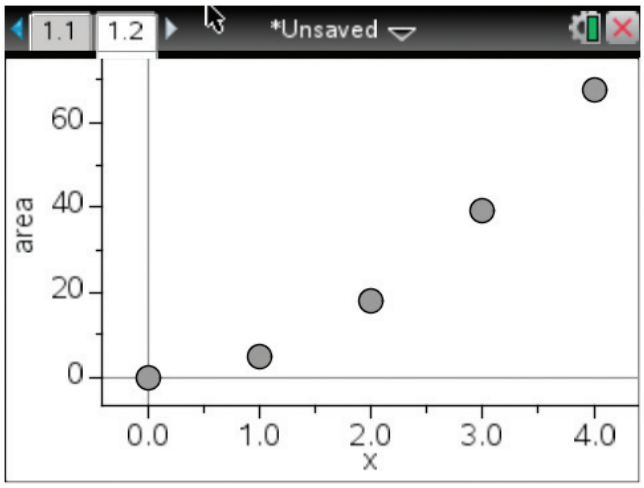
Area between $x = 0$ and $x = 4$ is $A_1 + A_2 + A_3 + A_4 = 5 + 13 + 21 + 29 = 68$ **[4 marks]**

Put the points (0, 0), (1, 5), (2, 18), (3, 39) and (4, 68) into a table.



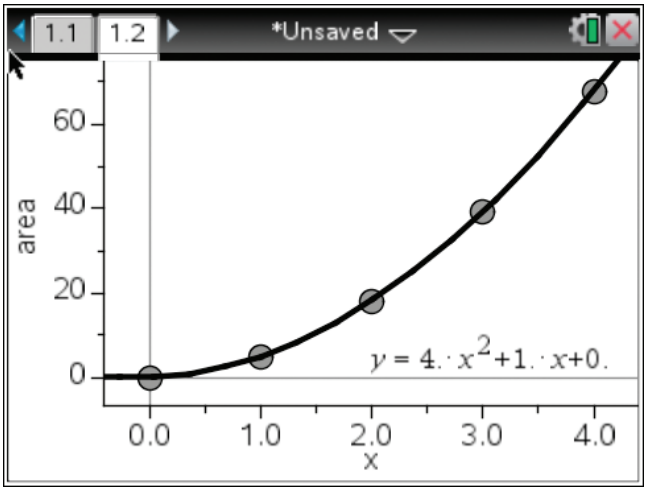
[1 mark]

Plot these points on a graph.



[1 mark]

Draw a quadratic regression.



$y = 4x^2 + x$ **[1 mark]**

4 Each rectangle has width 0.5 unit.

The heights are $f(0)$, $f(0.5)$, $f(1)$ and $f(1.5)$ **[1 mark]**

$A = 0.5 \times (0^2 + 1)^3 + 0.5 \times (0.5^2 + 1)^3 + 0.5 \times (1^2 + 1)^3 + 0.5 \times (1.5^2 + 1)^3$ **[1 mark]**

$= 22.64$ **[2 marks]**

5 a $-\int_3^1 e^x\,dx = \int_1^3 e^x\,dx$ **[1 mark]**

b $\int_1^2 (2x^4 - 3x^3 + x^2 - 3)dx + \int_1^2 (x^4 - x^3 + 5x^2 + 4x + 2)dx$

$= \int_1^2 (3x^4 - 4x^3 + 6x^2 + 4x - 1)dx$ **[1 mark]**

c $\int_{-3}^0 (2x^2 + 3)dx + \int_0^3 (2x^2 + 3)dx = \int_{-3}^3 (2x^2 + 3)dx$ **[1 mark]**

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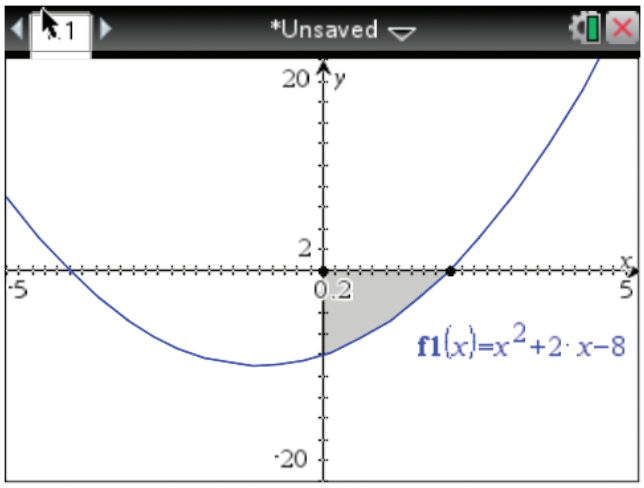
6 a $\int_0^2 (x^2 + 2x - 8)dx = \left[\frac{x^3}{3} + x^2 - 8x\right]_0^2$ **[1 mark]**

$= \left(\frac{2^3}{3} + 2^2 - 8 \times 2\right) - \left(\frac{0^3}{3} + 0^2 - 8 \times 0\right)$ **[1 mark]**

$= \frac{8}{3} + 4 - 16$

$= -9\frac{1}{3}$ **[1 mark]**

b



[1 mark]

The algebraic area between $x = 0$ and $x = 2$ is negative, so $-\int_0^2 (x^2 + 2x - 8)dx$ gives the area.

Total area $= -\int_0^2 (x^2 + 2x - 8)dx$ **[1 mark]**

$= -(-9\frac{1}{3})$

$= 9\frac{1}{3}$ **[1 mark]**

Part C – Analysis questions

1 a $\int_1^2 (y^2 - 1)dy = \left[\frac{y^3}{3} - y\right]_1^2$ **[1 mark]**

$= \left(\frac{2^3}{3} - 2\right) - \left(\frac{1^3}{3} - 1\right)$ **[1 mark]**

$= \frac{8}{3} - 2 - \frac{1}{3} + 1$

$= 1\frac{1}{3}$ **[1 mark]**

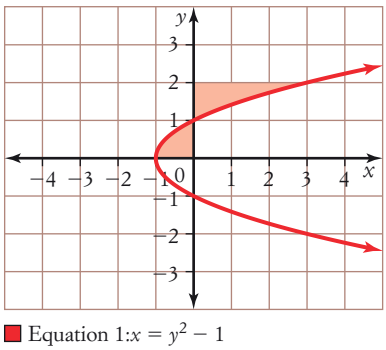
b $\int_0^1 (y^2 - 1)dy = \left[\frac{y^3}{3} - y\right]_0^1$ **[1 mark]**

$= \left(\frac{1^3}{3} - 1\right) - \left(\frac{0^3}{3} - 0\right)$ **[1 mark]**

$= \frac{1}{3} - 1$

$= -\frac{2}{3}$ **[1 mark]**

c



[2 marks]

d The algebraic area between $y = 0$ and $y = 1$ is negative, so $-\int_0^1 (y^2 - 1)dy$ gives the area.

Total area $= -\int_0^1 (y^2 - 1)dy + \int_1^2 (y^2 - 1)dy$ **[1 mark]**

$= -\left(-\frac{2}{3}\right) + 1\frac{1}{3}$ from parts **a** and **b** **[1 mark]**

$= 2$ units^2 **[1 mark]**

2 a $\int_0^3 50e^t\,dt = \left[50e^t\right]_0^3$ **[1 mark]**

$= 50e^3 - 50e^0$ **[1 mark]**

$= 50(e^3 - 1)$ **[1 mark]**

b $\int_4^9 50e^t\,dt = \left[50e^t\right]_4^9$ **[1 mark]**

$= 50e^9 - 50e^4$ **[1 mark]**

$= 50e^4(e^5 - 1)$ **[1 mark]**

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